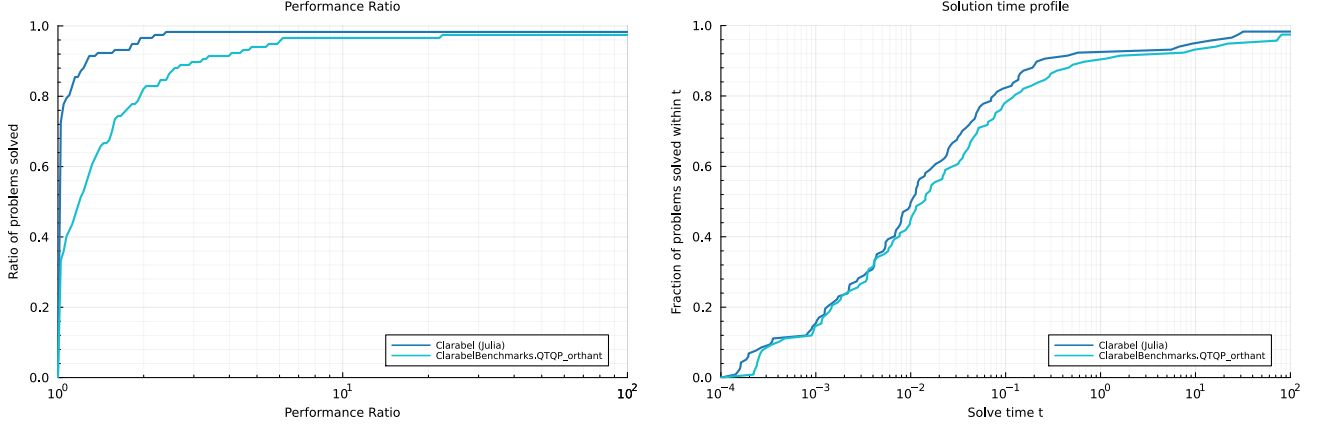


5 Numerical Experiments

5.1 Benchmark problems with quadratic objectives

5.1.1 The Maros-Meszaros test set

Figure 1: Performance profiles for the Maros-Meszaros problem set



(a) Relative performance profile

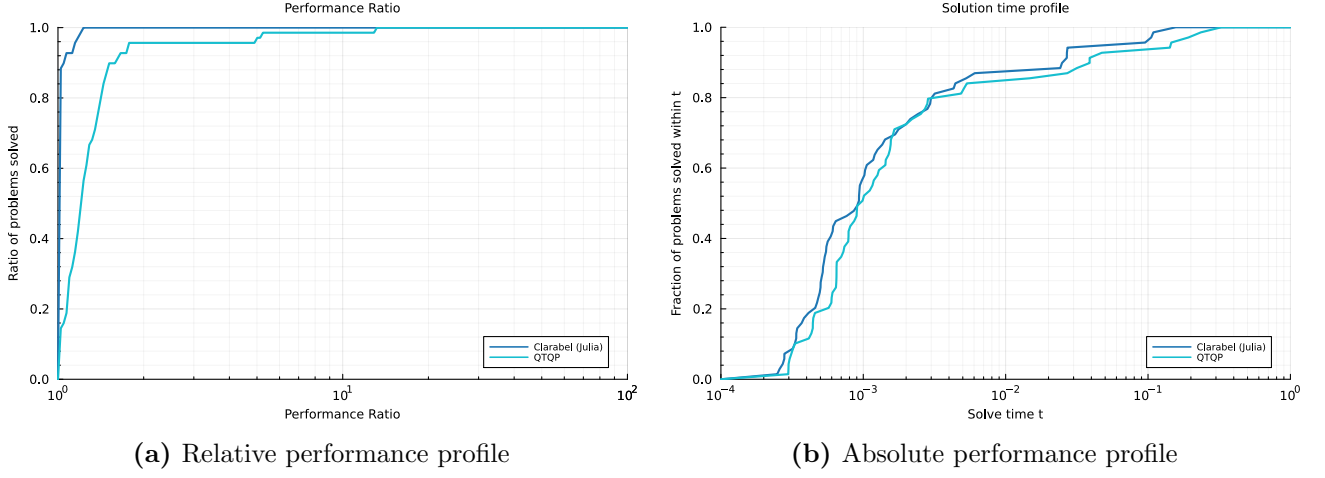
(b) Absolute performance profile

		Clarabel	ClarabelBenchmarks.QTQP_orthant
Shifted GM	Full Acc.	1.0	1.46
	Low Acc.	1.0	1.68
Failure Rate (%)	Full Acc.	1.7	2.6
	Low Acc.	0.9	2.6

(c) Benchmark timings as shifted geometric mean and failure rates

5.1.2 Constrained optimal control

Figure 2: Performance profiles for the optimal control problem set

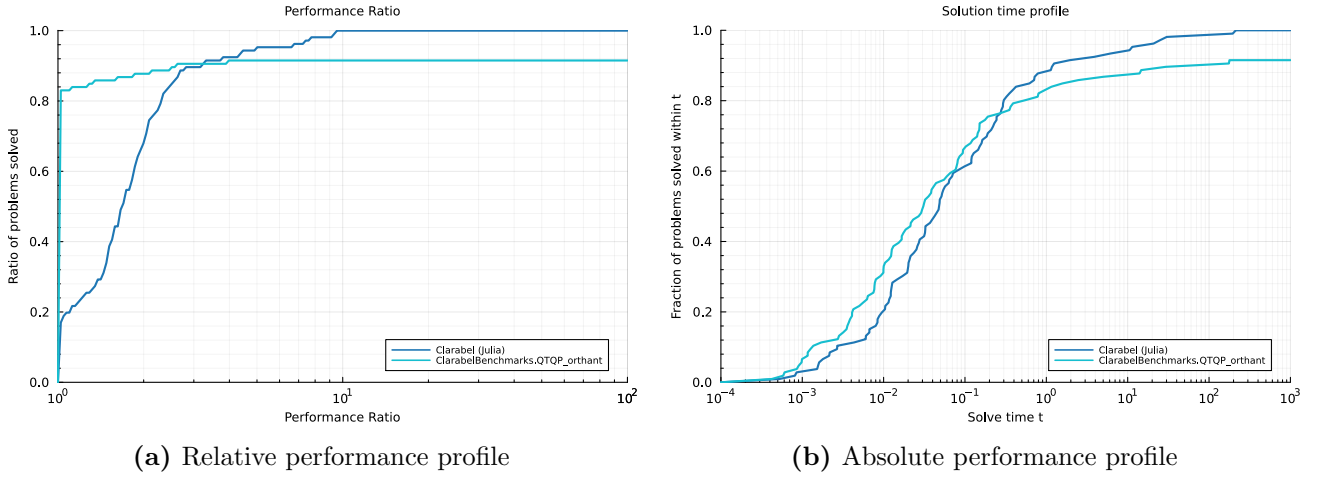


		Clarabel	QTQP
Shifted GM	Full Acc.	1.0	1.89
	Low Acc.	1.0	1.89
Failure Rate (%)	Full Acc.	0.0	0.0
	Low Acc.	0.0	0.0

(c) Benchmark timings as shifted geometric mean and failure rates

5.1.3 NETLIB LP problems

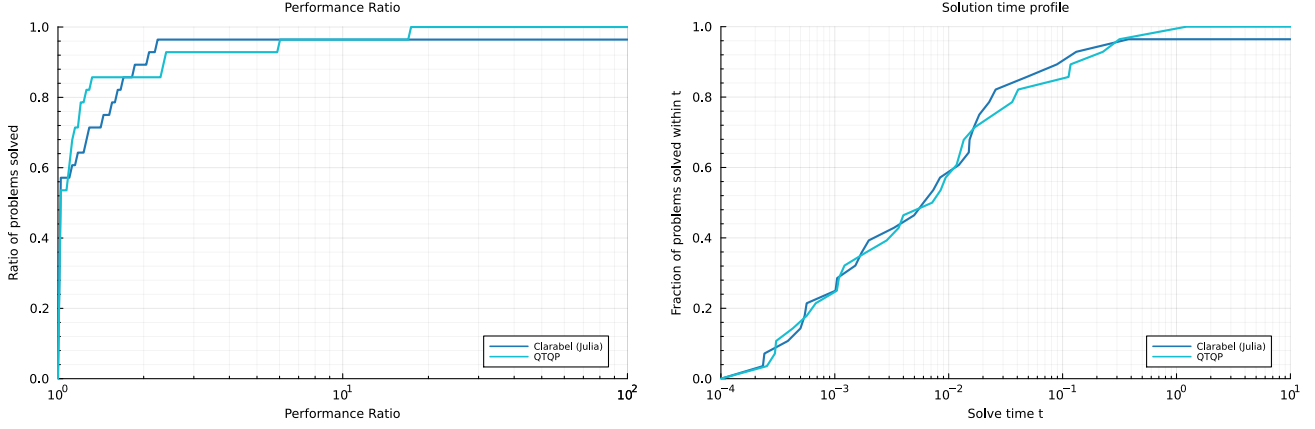
Figure 3: Performance profiles for the NETLIB Feasible LP problem set



		Clarabel	ClarabelBenchmarks.QTQP_orthant
Shifted GM	Full Acc.	1.0	2.45
	Low Acc.	1.0	2.45
Failure Rate (%)	Full Acc.	0.0	8.5
	Low Acc.	0.0	8.5

(c) Benchmark timings as shifted geometric mean and failure rates

Figure 4: Performance profiles for the NETLIB Infeasible LP problem set



(a) Relative performance profile

(b) Absolute performance profile

		Clarabel	QTQP
Shifted GM	Full Acc.	4.16	1.0
	Low Acc.	1.0	1.13
Failure Rate (%)	Full Acc.	3.6	0.0
	Low Acc.	0.0	0.0

(c) Benchmark timings as shifted geometric mean and failure rates

A Detailed benchmark results

Table 1: Solve times and iteration counts for the Quadratic programming problem set

	Problem	vars.	cons.	nnz(A)	nnz(P)	iterations		time per iteration(s)		total time (s)		
						Clarabel	QTQP	Clarabel	QTQP	Clarabel	QTQP	
ISSUE_103			5	7	10	15	7	7	3.41e-05	4.19e-05	0.000239	0.000293
HUBER_FITTING_N_10			55	45	87	15	7	6	3.84e-05	7.13e-05	0.000269	0.000428
PORTFOLIO_OPTIMIZATION_N_100			110	111	513	110	10	8	7.19e-05	9.29e-05	0.000719	0.000743
HUBER_FITTING_N_100			550	450	2642	150	8	8	0.000494	0.00108	0.00395	0.00861
SVM_L2_N_10			510	1000	1973	10	9	10	0.000283	0.000641	0.00255	0.00641
PORTFOLIO_OPTIMIZATION_N_1000			1100	1101	32065	1100	13	9	0.00405	0.00544	0.0526	0.049
HUBER_FITTING_N_500			2750	2250	50672	750	8	9	0.0276	0.0357	0.22	0.321
HUBER_FITTING_N_1000			5500	4500	195345	1500	9	10	0.243	0.176	2.19	1.76
PORTFOLIO_OPTIMIZATION_N_5000			5500	5501	759856	5500	18	11	0.209	0.254	3.76	2.8
SVM_L2_N_100			5100	10000	109929	100	11	13	0.0268	0.026	0.295	0.338
PORTFOLIO_OPTIMIZATION_N_10000			11000	11001	3019189	11000	18	12	2.37	2.39	42.6	28.7
PORTFOLIO_OPTIMIZATION_N_15000			16500	16501	6782628	16500	18	12	5.79	5.55	104	66.6
SVM_L2_N_500			25500	50000	2548936	500	12	14	1.13	1.07	13.6	14.9